An Exact Centered-Star Repair and Its Entropy Cost

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Abstract

We construct the natural radial repair on the fifteen-dimensional centered tangent space of a six-staple $SU(2)$ star and retain its complete Jacobian entropy.

For $x = (x_1, \dots, x_m) \in (\mathbb{R}^c)^m$, write $\bar{x} = m^{-1} \sum_i x_i$ and define

$$T_a(x)_i = \bar{x} + a(x_i - \bar{x}), \qquad 0 < a < 1.$$

Theorem 1. *The map preserves the mean, is equivariant under common orthogonal adjoint action, multiplies centered dispersion by a^2, and has determinant $a^{c(m-1)}$ on the centered fiber. If the bad RMS is at least η, then for Gaussian weight $e^{-D/(2t)}$ its ideal repair free energy is*

$$F_{\mathrm{star}} = \frac{(1-a^2)m\eta^2}{2t} - c(m-1)\log(1/a).$$

Proof. The mean and scaling identities follow by direct substitution. The centered subspace has dimension $c(m-1)$, giving the determinant. The action difference is at least $(1-a^2)m\eta^2/(2t)$; change of variables supplies the Jacobian loss. □

The frozen six-staple, three-color tangent repair exceeds the allocated free-energy target even after its fifteen-dimensional Jacobian is charged.

Remark 1. *This map acts on independently represented staple coordinates. A lift to actual link variables may create curvature outside the star and is not asserted here.*